

IPC144 – Introduction to C

Programming

Practice Exercises

Week 10 - C Formatted

Input/Output

9.7 (Differences Between %d and %i) Write a program to test the difference between the %d and %i conversion specifiers when used in scanf statements. Ask the user to enter two integers separated by a space. Use the statements

```
scanf("%i%d", &x, &y);  
printf("%d %d\n", x, y);
```

to input and print the values. Test the program with the following sets of input data:

```
10 10  
-10 -10  
010 010  
0x10 0x10
```

9.8 (Printing Numbers in Various Field Widths) Write a program to test the results of printing the integer value 12345 and the floating-point value 1.2345 in fields of various sizes. What happens when the values are printed in fields containing fewer digits than the values?

9.9 (Rounding Floating-Point Numbers) Write a program that prints the value 100.453627 rounded to the nearest digit, tenth, hundredth, thousandth and ten-thousandth.

9.10 (Temperature Conversions) Write a program that converts integer Fahrenheit temperatures from 0 to 212 degrees to floating-point Celsius temperatures with 3 digits of precision. Perform the calculation using the formula

$$\text{celsius} = 5.0 / 9.0 * (\text{fahrenheit} - 32);$$

The output should be printed in two right-justified columns of 10 characters each, and the Celsius temperatures should be preceded by a sign for both positive and negative values.

9.11 (Escape Sequences) Write a program to test the escape sequences '\', '\', '\?', '\\, \a, \b, \n, \r and \t. For the escape sequences that move the cursor, print a character before and after printing the escape sequence so it's clear where the cursor has moved.

9.12 (Printing a Question Mark) Write a program that determines whether ? can be printed as part of a printf format control string as a literal character rather than using the \? Escape sequence.

9.13 (Reading an Integer with Each scanf Conversion Specifier) Write a program that inputs the value 437 using each of the scanf integer conversion specifiers. Print each input value using all the integer conversion specifiers.

9.14 (Outputting a Number with the Floating-Point Conversion Specifiers) Write a program that uses each of the conversion specifiers e, f and g to input the value 1.2345. Print the values of each variable to prove that each conversion specifier can be used to input this same value.

9.15 (Reading Strings in Quotes) In some programming languages, strings are entered surrounded by either single or double quotation marks. Write a program that reads the three strings suzy, "suzy" and 'suzy'. Are the single and double quotes ignored by C or read as part of the string?

9.16 (Printing a Question Mark as a Character Constant) Write a program that determines whether? can be printed as the character constant '?' rather than the character constant escape sequence '\?' using conversion specifier %c in the format control string of a printf statement.

9.17 (Using %g with Various Precisions) Write a program that uses the conversion specifier g to output the value 9876.12345. Print the value with precisions ranging from 1 to 9.